SOLUTION

SECTION–I

1. Ans. (D)

Sol. $(1 + z)^6 + z^6 = 0$

$(1 + z)^6 = -z^6$

$$\left(\frac{z+1}{z}\right)^6 = -1$$

$$\left|\frac{z+1}{z}\right|^6 = 1$$

$|z + 1| = |z|$

Thus : z lies on perpendicular bisector of line segment joining (–1,0) and (0,0)

Hence : all roots lie on line and arc collinear.

2. Ans. (B)

$$P = \cos\frac{2\pi}{7} + i\sin\frac{2\pi}{7}$$

$$P = e^{\frac{2\pi}{7}i} \Rightarrow P^7 = \left(e^{\frac{2\pi}{7}i}\right) = e^{2\pi i}$$

$a + b = P + P^2 + P^4 + P^3 + P^5 + P^6$

$a + b + 1 = 1 + P + P^2 + ... + P^6$

$$= \frac{P^7 - 1}{P - 1}\{P^7 = 1\}$$

$a + b + 1 = 0$

$\therefore a + b = -1$

$ab = (P + P^2 + P^4)(P^3 + P^5 + P^6)$

$= P^4 + P^6 + P^7 + P^5 + P^7 + P^8 + P^7 + P^9 + P^{10}$

$= P^4 + P^6 + P^5 + P^8 + P^9 + P^{10} + 3$

$P^4 + P^5 + P^6 + P^7(P + P^2 + P^3) + 3$

$P + P^2 + ... + P^6 + 3$

$-1 + 3 = 2$

$\therefore |(a + b) + abi|$

$$= |-1 + 2i| = \sqrt{(-1)^2 + (2)^2} = \sqrt{5}$$

3. Ans. (D)

Sol. since $\text{Im}(z_1) + \text{Im}(z_2) + \text{Im}(z_3) = \text{Im}(z_1 + z_2 + z_3)$

$= \text{Im}(\text{sum of roots})$

$= \text{Im}(1 + 3i) = 3$

Alternate :

$(z^3 - z^2 - 2z + 2) + 3i(z - z^2) = 0$

$(z - 1)(z^2 - 2) + 3iz(1 - z) = 0$

$(z - 1)(z^2 - 2 - 3iz) = 0$

$z = 1,\ z = 2i, z = i$

$\text{Im}(z_1) + \text{Im}(z_2) + \text{Im}(z_3) = 3.$

4. Ans. (B)

Sol. $|z - 2i| = 2$ is a circle with centre 2i

$z(1 - i) - \bar{z}(1 + i) = 4i$ is a line

This line passes through center of circle.

$\therefore$ Total 2 points of intersection.

5. Ans. (D)

Sol. $$\sum_{k=1}^{100} i^{k!} + \sum_{k=1}^{100} \omega^{k!}$$

$$\sum_{k=1}^{100} i^{k!} = i^{1!} + i^{2!} + i^{3!} + i^{4!} + ... + i^{100!}$$

$= i - 1 + i^6 + 1 + 1 + 1 + ... + 1 = i - 2 + 97$

$= i + 95.$

$$\sum_{k=1}^{100} \omega^{k!} = \omega^1 + \omega^{2!} + \omega^{3!} + \omega^{4!} + ... + \omega^{100!}$$

$= \omega + \omega^2 + 1 + 1 + 1 + ... + 1 = 97$

sum $= i + 95 + 97 = i + 192$

6. Ans. (D)

Sol. $$\arg\left(\frac{z_1}{z_4}\right) + \arg\left(\frac{z_2}{z_3}\right) = \arg\left(\frac{z_1 z_2}{z_3 z_4}\right)$$

$$\arg\left(\frac{z_1\bar{z}_1}{z_3\bar{z}_3}\right) = \arg\left(\frac{|z_1|^2}{|z_3|^2}\right) = 0$$

7. Ans. (D)

Sol. $z = x + iy$

$|(x - 3) + (y - 2)i| = |x + (y + 2)i|$

$(x - 3)^2 + (y - 2)^2 = x^2 + (y + 2)^2$

$6x + 8y = 9$

$|z|$ means distance from origin

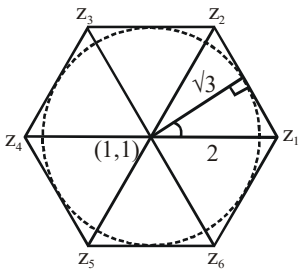
$$\text{So } |z|_{\text{min.}} = \left|\frac{0 + 0 - 9}{\sqrt{6^2 + 8^2}}\right| = \frac{9}{10}$$

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8. Ans. (C)

Sol.



$\left(2+\left(1+\sqrt{3}\right)i\right)$

$z_2-(1+i)=\left(2+\left(1+\sqrt{3}\right)i-(1+i)\right)e^{i\pi/3}$

$\Rightarrow \quad z_2-(1+i)=\left(1+i\sqrt{3}\right)e^{i\pi/3}$

$=\left(1+i\sqrt{3}\right)\left(\frac{1+\sqrt{3}i}{2}\right)=-1+\sqrt{3}i$

$\Rightarrow z_2=\left(1+\sqrt{3}\right)i$

similarly using concept of rotation find z_6 and z_5

$\Rightarrow z_6=3+i$

$z_5=2+\left(1-\sqrt{3}\right)i$

9. Ans.(A,B,C,D)

Sol. $|2z_1-3z_2-4z_3|=|2z_1-3(z_2-2)-4(z_3-5)-26|$

$\le 2|z_1|+3|z_2-2|+4|z_3-5|+|26|\le 2+9+24+26\le 61$

10. Ans.(B,D)

Sol. $zw+w^2=\bar{z}$

$1+zw=w^2\bar{z}$

$\Rightarrow \quad \bar{z}-w^2=\bar{z}w^2-1 \Rightarrow (\bar{z}+1)(1-w^2)=0$

$\Rightarrow \quad \bar{z}=-1$ or $w^2=1$

Case I : $\bar{z}=-1 \Rightarrow \arg z=\pi$

$\Rightarrow \quad -1+w=-\frac{1}{w} \Rightarrow w^2-w+1=0$

$\Rightarrow \quad \arg(w)=-\frac{\pi}{3}$ or $\frac{\pi}{3}$

Case II : $w=-1$

$\Rightarrow \quad z-1=-\bar{z} \quad \Rightarrow \quad \text{Re}(z)=\frac{1}{2}$

11. Ans. (B,C)

Sol. $\because \ |z_3|=\frac{|z_2|.|z_1-z_4|}{|\bar{z}_1z_4-1|}=\frac{|z_1-z_4|}{|\bar{z}_1z_4-1|}\le 1$

$\Rightarrow |z_1-z_4|^2\le|\bar{z}_1z_4-1|^2$

$\Rightarrow (z_1-z_4)(\bar{z}_1-\bar{z}_4)\le(\bar{z}_1z_4-1).(z_1\bar{z}_4-1)$

$\Rightarrow |z_1|^2+|z_4|^2-|z_1|^2|z_4|^2-1\le 0 \Rightarrow (|z_1|^2-1)(|z_4|^2-1)\ge 0$

$\because |z_1|<1 \Rightarrow |z_4|\le 1$

Paragraph for Question 12 to 14

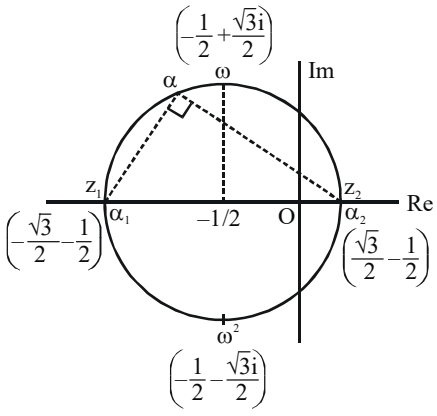
$(z-\omega)^2+(z-\omega^2)^2=0$

$\Rightarrow \quad 2z^2-2z(\omega+\omega^2)+\omega^4+\omega^2=0$

$\Rightarrow \quad z_1=-\frac{\sqrt{3}}{2}-\frac{1}{2}i;\ z_2=\frac{\sqrt{3}}{2}-\frac{1}{2}i$

α lies on circle with centre $-\frac{1}{2}$ and radius $\frac{\sqrt{3}}{2}$

$\alpha_1=-\frac{\sqrt{3}}{2}-\frac{1}{2}i;\ \alpha_2=\frac{\sqrt{3}}{2}-\frac{1}{2}i$



12. Ans. (C)

Sol. z_1, z_2 and ω, ω^2 are diametrical ends of given circle

$\therefore \quad |z_1-\alpha|^2+|z_2-\alpha|^2+|\omega-\alpha|^2+|\omega^2-\alpha|^2$

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$= (\sqrt{3})^2 + (\sqrt{3})^2 = 6$

13. Ans. (B)

Sol. $\alpha_1 = z_1$ and $\alpha_2 = z_2$

$\therefore$ (B) is correct.

14. Ans. (C)

Sol. $|\alpha - \alpha_1| + |\alpha - \alpha_2| = \sqrt{6}$

Here α moves on an ellipse having its focii as α_1 and α_2 and length of minor axis as $\sqrt{3}$ units.

$\therefore$ 2 points for $|\alpha - \alpha_1| + |\alpha - \alpha_2| = \sqrt{6}$

SECTION–III

1. Ans. 4.50

Sol. Let $z = x + iy$

$\Rightarrow |x - 2 + iy| = 3|x + iy + 2|$

$\Rightarrow (x-2)^2 + y^2 = 9(x+2)^2 + 9y^2$

$\Rightarrow 8x^2 + 8y^2 + 40x + 32 = 0$

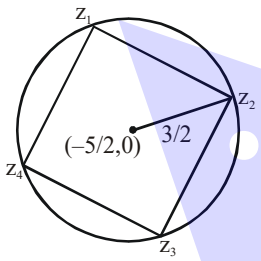
$\Rightarrow x^2 + y^2 + 5x + 4 = 0$

Clearly locus is a circle with centre $\left(-\frac{5}{2}, 0\right)$ and radius

$= \sqrt{\frac{24}{4} - 4} = \frac{3}{2}$

since $z_1, z_2, z_3, z_4 \in A$

$\Rightarrow$ all these points must lie on this circle and forms a parallelogram, since $z_1 + z_3 = z_2 + z_4$



Now to have maximum area of parallelogram, it must be a square whose area will be

$\frac{d^2}{2} = \frac{9}{2}$

2. Ans. 1.00

Sol. $\alpha = \frac{1 \pm \sqrt{3}\,i}{2} \Rightarrow \alpha = -\omega \text{ or } -\omega^2$

$\therefore \quad (-\omega)^{333} + (-\omega)^{666} + (-\omega)^{999} = -1 + 1 - 1 = -1$

3. Ans. 5.00

Sol. $z_1 = 4 + 3i$

After moving $\sqrt{2}$ units on the direction of $-\hat{i} + \hat{j}$

$z_2 = (4 - 1) + (3 + 1)\,\hat{i}$

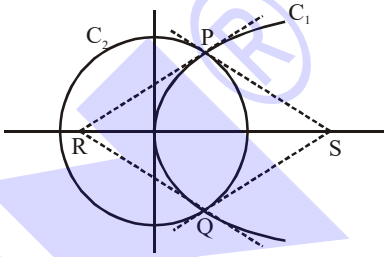
$= 3 + 4i$

z_2 is rotated through an angle $\frac{\pi}{2}$ in clockwise direction

so $p + iq = (3 + 4i)\, e^{-i\frac{\pi}{2}} = (3 + 4i)(-i) = 4 - 3i$

4. Ans. 100.00

Sol.



Put $z = x + iy$

$C_1 : y^2 = 8x$

$C_2 : x^2 + y^2 = 9$

On solving C_1 & C_2

$P(1, 2\sqrt{2})$

Tangent to C_1 at P : $y(2\sqrt{2}) = 8\frac{(x+1)}{2}$

at $y = 0 : x = -1 \Rightarrow R(-1, 0)$

Tangent to C_2 at P : $x(1) + y(2\sqrt{2}) = 9$

at $y = 0 : x = 9 \Rightarrow S(9, 0)$.

$\therefore$ Area $\Delta PRS : \frac{1}{2} \times (9 - (-1)) \times 2\sqrt{2} = 10\sqrt{2}$

$\therefore \quad \lambda = 10 \Rightarrow \lambda^2 = 100$

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